



LiRI Corpus Platform: Demonstration of a Web-Based Infrastructure for Multimodal Corpus Analysis

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Abstract

We present the LiRI Corpus Platform (LCP), a web-based infrastructure for storing, exploring, and analyzing diverse linguistic corpora with a focus on multimodal and audiovisual data. The platform supports synchronized querying of text, speech, and video through unified interfaces and a custom query language (Descriptive Query Definition - DQD). Dedicated frontends enable time-aligned exploration of gesture, audio, and spoken transcripts. LCP is designed to support researchers working with multimodal and annotated datasets, enabling cross-modal queries and layered annotation.¹

Index Terms: corpus platform, multimodal corpora, audiovisual corpora, information retrieval, query language

1. Introduction

Multimodal corpora combining text, speech, and video are increasingly central to research in linguistics, interaction analysis, and digital humanities. However, tools that allow integrated access to these data types remain limited. Most existing platforms are either text-centric or restricted to audiovisual annotation without query support [1, 2, 3, 4].

Researchers often rely on a patchwork of tools to manage multimodal data, making workflows fragmented and inefficient. Some solutions, such as ANNIS [5] or CQPweb [6], provide powerful textual querying but lack integrated audiovisual support. Others like VIAN [7] focus on rich video annotation but do not offer full corpus query capabilities.

The LiRI Corpus Platform (LCP) addresses this gap by offering an integrated solution for corpus storage, exploration, and analysis across modalities, developed at the University of Zurich as part of the LiRI (Linguistic Research Infrastructure) [8]. LCP is the only platform including web-based querying, data upload, and multi-modal analysis. It supports corpus-based studies in linguistics, sociolinguistics, interaction research, media and communication studies, and the digital humanities, but it is also useful for broader fields such as human-computer interaction, behavioral science, etc.

LCP builds on previous experience in managing multilingual, richly annotated corpora and aims to provide a user-friendly, extensible solution for both academic and applied research, emphasizing flexibility, transparency, and interoperability [9, 10].

2. System Architecture

LCP features a modular, service-oriented architecture centered around a PostgreSQL backend. Data is structured using a flexi-

ble entity-relation model, defined per corpus via a user-supplied template. Each template describes entities (e.g., tokens, gestures, turns, scenes) and their alignment to one or more dimensions:

- **Symbol axis:** character positions in text
- **Time axis:** milliseconds in audio or video
- **Space axis:** regions in images (planned)

This model allows corpora to include overlapping spans, multimodal alignments, and nested annotations without structural constraints. All data is accessible through a unified query language, Descriptive Query Definition (DQD), which supports entity pattern matching, logical filtering, and result aggregation (e.g., frequency lists, collocation tables).

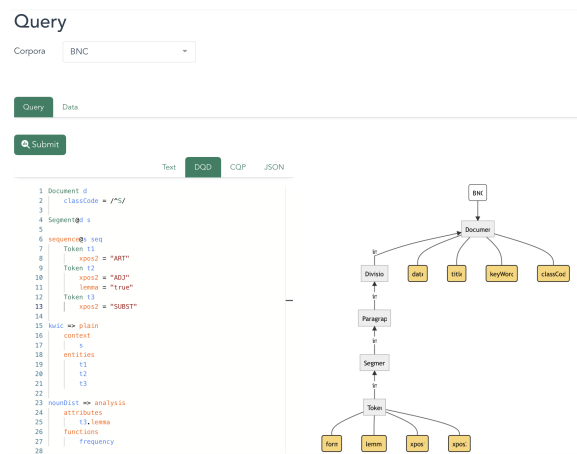


Figure 1: Query interface with the corpus template representing the British National Corpus [11]

3. Query Language: Descriptive Query Definition (DQD)

The DQD language provides a declarative formalism to query heterogeneous corpus data in LCP (Fig.1, left). Designed for flexibility and expressiveness, DQD queries operate over the user-defined corpus template (Fig.1, right) and support multi-level, cross-modal search operations. Users can define entities, sequences, filters, and constraints using a readable syntax inspired by natural language descriptions.

Crucially, DQD supports querying across modalities and annotation layers. For example, users can search for spoken utterances by specific speakers that co-occur with gesture annotations in a given time window, or retrieve all noun phrases

¹ Acknowledgements: This project was supported by University of Zurich, swissuniversities, DIZH, CLARIN-CH, SwissUBase.

aligned with a speaker turn in a particular metadata category (e.g., age group, region).

Queries can return three main result types:

- **Plain:** concordance-style results with optional context.
- **Analysis:** frequency or distributional statistics.
- **Collocation:** statistical co-occurrence results within a span.

The language supports logical operators, quantifiers, negation, and optional constraints on temporal alignment, speaker identity, or metadata fields [8]. The result interface highlights matching spans and supports download and export for further analysis.

4. Specialized Interfaces

Three frontends offer tailored access to different modalities:

- **Catchphrase** — text corpora: supports linguistic querying, collocation analysis, and metadata filtering.
- **SoundScript** — speech corpora: displays speaker segments, aligned transcripts, and audio waveforms with time-based search.
- **VideoScope** — audiovisual corpora: enables querying and browsing of layered video annotations such as gestures, head nods, and facial expressions (Fig. 2).

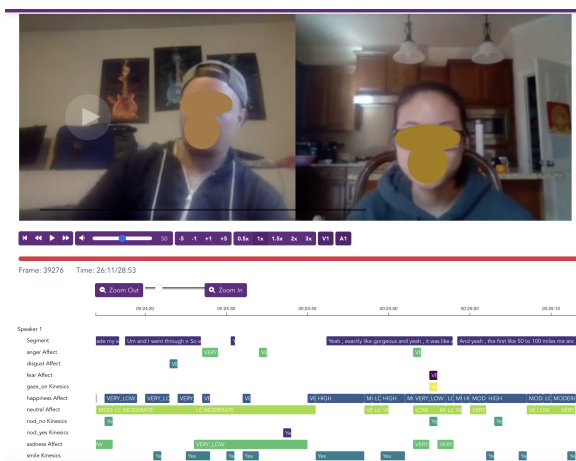


Figure 2: VideoScope interface with the timeline showing the multimodal CANDOR corpus[12]

All frontends offer visual navigation, interactive result exploration, and export options. DQD queries are validated automatically based on the corpus template, and executed with incremental sampling to ensure fast response times on large datasets.

5. Demonstration and Use Cases

We will demonstrate several corpora live in the LCP interface:

- **CANDOR corpus**, a richly annotated multimodal corpus of interactions [12], showing synchrony between visual information and speech in dyadic conversations in VideoScope.
- **OFROM corpus**[13], a spoken corpus of Swiss French speakers, illustrating how SoundScript supports spoken language analysis.

We will show how users import their own data, define templates, run complex multimodal queries, and export results.

6. Conclusion

The LiRI Corpus Platform provides an end-to-end solution for working with multimodal corpora, combining flexible data modeling, expressive querying, and modality-aware user interfaces. It supports advanced research in speech, interaction, and audiovisual communication by bridging the gap between annotation tools and scalable analysis. The platform is open to collaboration and is used in several interdisciplinary projects at the University of Zurich and beyond.

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